

Name: Saiprasad Ashokrao Kawdikar

PRN No: 202501040030

3.2.5. Numpy: Copying and Viewing Arrays

04:17

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: **<original_array >**

View array: **<view_array >**

- After modifying the copy:

Original array after modifying copy: **<original_array >**

Copy array: **<copy_array >**

```
import numpy as np
```

```
# Input array from the user
array1 = np.array(list(map(int, input().split()))))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))
# Find indices where value matches in array1
indices = np.where(array1 == search_value)[0]
print(indices)

# Count occurrences in array1
count = np.count_nonzero(array1 == count_value)
print(count)

# Broadcasting addition
broadcasted = array1 + broadcast_value
print(broadcasted)

# Sort the first array
sorted_array = np.sort(array1)
```

```
print(sorted_array)
```

Average time
0.017 s
16.50 ms

Maximum time
0.018 s
18.00 ms

✓ 2 out of 2 shown test case(s) passed
✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 18 ms Debug

Expected output	Actual output
1 1 1 2 2 2	1 1 1 2 2 2
value to search: 1	value to search: 1
value to count: 2	value to count: 2
value to add: 2	value to add: 2
[0, 1, 2]	[0, 1, 2]
3	3
[3, 3, 3, 4, 4, 4]	[3, 3, 3, 4, 4, 4]
[1, 1, 1, 2, 2, 2]	[1, 1, 1, 2, 2, 2]

✓ Test case 2 15 ms